Supplementary Material: When Agreement Hides Error

Anonymous CVPR submission

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1. Scope and Evidence Contract

This supplement reports protocol details, full strata, and every result that materially limits a submitted claim. Experiments are included according to relevance, not favorability. The manuscript does not claim a new reconstruction architecture, calibrated uncertainty, universal estimator superiority, plane-invariant influence, local failure localization, or prospective acquisition benefit.

The independent units are 24 fastMRI examinations, 115 IXI examinations, and eight NingXia patients containing 22 series. All NingXia intervals resample patients with subordinate series, planes, models, and contexts retained. Dense target slices are used only for retrospective evaluation and never in Anchor-CV, adjacent-anchor, disagreement, or influence scores.

2. Definitions and Frozen Protocol

For predictions $f_a = y + e_a$ and $f_b = y + e_b$, shared and observable error fractions are

$$S = \frac{\|\bar{e}\|^2}{\|\bar{e}\|^2 + \|\delta\|^2}, \qquad O = \frac{\|\delta\|^2}{\|\bar{e}\|^2 + \|\delta\|^2}, \tag{1}$$

where $\bar{e} = (e_a + e_b)/2$ and $\delta = (e_a - e_b)/2$. Numerical identity error is below 4.42×10^{-10} in the stored audit.

For nonlinear models, four frozen seeds (6101–6104) generate centered Rademacher fields at 1% of the globally normalized case range. Each shared anchor is perturbed separately. The response-cosine statistic averages identical-field response cosines across shared anchors and seeds. The broad grid uses reference offsets $(-2, +2)$ and eight target-preserving comparators. Span and asymmetry are explicit covariates; the complete offset list is stored with the experiment registry. The fixed-count subset contains the four comparators with exactly one common anchor.

3. Plan-9 Architecture-Family Protocol

The architecture hypothesis and falsification conditions were frozen before new P0 results. Phase-0 anchors are $\{0, 4, 8, 12\}$, phase-2 anchors are $\{2, 6, 10, 14\}$, and common missing targets are $\{3, 5, 7, 9, 11\}$. The clean shared control applies an in-plane horizontal flip to phase-0 inputs and maps outputs back. A one-case smoke using through-plane reversal was rejected before cohort inference because it changed physical geometry and exposed direction non-equivariance; its outputs are retained but excluded. Regenerated phase-0 outputs match stored predictions exactly for every one of 24 cases and all five architectures (pixel MAE and maximum absolute difference zero).

I3Net uses four regular anchors and outputs 13 planes. SAINT consumes a left/right pair plus relative and absolute position maps. The internal CycleINR-style model linearly densifies four regular anchors and queries a coordinate-conditioned residual network. The adapted SAINT and CycleINR-style implementations are not claimed as exact official reproductions. All interventions occur at physical anchor identity; no architecture is distorted into another input interface.

The equal-weight four-family observable gain is .155 [.148,.162]. Leave-one-family-out estimates are .180 without I3Net, .131 without INR, .161 without SAINT, and .147 without transformers; all intervals exclude zero. The frozen shared-condition thresholds identify stable-wrong rates of 0 for HAT, SwinIR, I3Net, and SAINT and 4/120 (3.33%) for CycleINR; CycleINR has 0/120 under disjoint contexts. This negative prevalence result is not replaced by the earlier Plan-8 definition.

4. New-Architecture Influence Stability and Boundary

SAINT uses all 24 fastMRI examinations and three deterministic interior planes. I3Net/CycleINR use the first 24 geometry-eligible cases in frozen IXI test-list order and one central plane. Their regular four-anchor reference is $(-5, -1, +3, +7)$, with two overlap-three and five zero-overlap contexts. Zero-overlap influence is zero by physical-measurement definition and is excluded from the stability gate.

SAINT's broad influence associations have the expected direction for alignment, shared fraction, observable fraction, and disagreement. Its fixed-count estimates also have the expected signs, but only disagreement excludes

Table 1. Architecture-family same-target result on 24 fastMRI examinations at $\times 4$. Observable gain is disjoint minus clean shared fraction. Ranking intervals resample examinations.

Model	Family	Observable gain [95% CI]	$\rho_{\text{shared}} \rightarrow \rho_{\text{disjoint}}$	Ranking $\Delta\rho$ [95% CI]
HAT	Transformer SR	.178 [.170,.187]	.896→.952	.057 [−.030,.186]
SwinIR	Transformer SR	.176 [.164,.188]	.883→.945	.062 [−.051,.234]
I3Net	Inter/intra-slice CNN	.080 [.072,.088]	.943→.961	.018 [−.061,.107]
SAINT	Spatial conditional CNN	.137 [.125,.149]	.934→.924	−.010 [−.143,.114]
CycleINR	Implicit coordinate	.226 [.219,.233]	.634→.976	.342 [.113,.711]

Table 2. Plan-9 influence audit. Stability entries are rank correlations with the $+1\%$ primary statistic. A model is claim eligible only if all are at least .90.

Model	$+.5\%$	$+2\%$	-1%	Broad $\rho(I,\text{alignment})$	$\lvert\rho_I\rvert - \lvert\rho_{count}\rvert$ [CI]	Fixed-count partial ρ [CI]
SAINT	.940	.964	.995	.706 [.680,.734]	.014 [−.002,.029]	.063 [−.021,.149]
I3Net	.859	.737	.939	.776†	−.006†	.434†
CycleINR	.779	.857	.876	.789†	.007†	.229†

†Exploratory only after the preregistered stability gate failed; excluded from influence claims.

zero. This supports a stable local influence signal in a non-transformer while falsifying a universal beyond-count advantage. I3Net/CycleINR remain valid phenomenon experiments because their controlled predictions do not depend on the finite-response stability classification.

5. Controlled Dependence Results

The original order-reversal control is not used as the clean causal contrast. It supports the complete covariance chain on IXI and NingXia, but several cosine/shared/observable arrows reverse on fastMRI because order reversal changes more than anchor dependence. The horizontal-flip clean control preserves anchor identities, target, distances, count, z order, mask, and preprocessing and is therefore the primary intervention.

Metric robustness was assessed over MAE, RMSE/MSE, SSIM error, gradient MAE, low-pass MAE, and residual high-frequency MAE for three physical planes, three cohorts, and two architectures. All 42 disjoint-minus-shared point differences are positive. All seven IXI intervals exclude zero. SSIM error has weak or negative absolute ranking on fastMRI/NingXia and several smaller-cohort gradient/high-frequency intervals cross zero. The claim is broad error observability, not metric independence.

6. Exact Count Falsification and Analytical Influence

The compact analytical ladder uses all valid sign configurations at fixed absolute distances $\{1, 2, 3, 4\}$. Across fastMRI, IXI, and NingXia, overlap count has overall $+.9$ association with error cosine/shared fraction and $-.9$ to -1.0 with observable fraction/disagreement. Every cohort nevertheless shows a zero-to-one reversal. A wider $\{2, 4, 6, 8\}$ ladder reproduces the exception on IXI while NingXia is monotone. This rejects strict count monotonicity without rejecting a broad dependence effect.

For the four main analytical endpoints, clustered intervals for the absolute-correlation improvement of common coefficient mass over count are all above zero, spanning .189–.283. Signed dot product and coefficient cosine agree qualitatively; common mass is reported because it retains unequal shared magnitude.

7. Full Neural Shared-Influence Results

The corresponding primary shared-fraction and observable-fraction advantages are .040/.040 and .056/.056 on fastMRI, .089/.089 and .081/.081 on IXI, and .049/.049 and .043/.043 on NingXia for HAT/SwinIR. Every interval excludes zero. Direct disagreement improves in all six strata, but both fastMRI intervals cross zero; it is supporting rather than primary evidence.

7.1. Fixed-one-overlap test

NingXia does not preserve the expected direction consistently and all intervals cross zero. We therefore do not use wording such as "the direction is preserved but less precise." The actual result is null/mixed under eight-patient fixed-count inference.

7.2. Perturbation and plane stability

The Plan-7 audit varies amplitude (.5/1/2%), positive/negative field, Rademacher/Gaussian field, and 1/2/4/8 directions. All 42 cohort–architecture–condition strata pass

Table 3. Clean same-target dependence control. Stable-wrong Δ uses high-error/low-disagreement thresholds frozen from the zero-overlap condition.

Cohort	Units	ρ_{disjoint}	ρ_{shared}	$\Delta\rho$ [95% CI]	Shared / observable error
fastMRI	24 examinations	.959	.867	.092 [.043,.176]	97.39% / 2.61%
IXI	115 examinations	.981	.886	.095 [.078,.126]	98.42% / 1.58%
NingXia	8 patients, 22 series	.892	.698	.193 [.111,.371]	99.75% / .25%

Table 4. Analytical shared influence versus count. Values are Spearman association with error alignment.

Cohort	Count	Common coefficient mass	Gain
fastMRI	.503	.815	.312
IXI	.466	.789	.323
NingXia	.630	.855	.225

the registered rank-stability threshold; minimum Spearman is .942 and maximum .999. Physical-plane pair stability is .779–.954, and only one of three plane-pair gates passes. This makes the statistic robust to the tested numerical perturbation design but local to a declared physical plane.

8. Mechanism Chain

The context-ranking endpoint uses only eight configurations and is not a mediation estimate. Influence-to-ranking is still less stable. The manuscript therefore claims that influence is one determinant of local error alignment/observability, not that it alone determines global case ranking.

9. Anchor-CV, Strong Baselines, and Consequence

Anchor-CV exactly equals the masked MAE of half the acquired-slice second difference. Frozen-score reproduction has maximum absolute error 5.68×10^{-14}. Table 8 reports strong baselines that prevent a uniqueness claim.

Partial rank association for Anchor-CV after adjacent MAE, robust scale, and spacing is .333 on fastMRI, $-.059$ on IXI, and $-.100$ on NingXia. Scale normalization reduces/reverses rankings, so the score is not a scale-free invariant. Frozen risk–coverage also has no universal winner: output PC is better on fastMRI/IXI and adjacent MAE is better on NingXia.

For $\times 4 \rightarrow \times 2$ benefit, Anchor-CV/adjacent MAE/output PC Spearman is .871/.846/.843 on fastMRI and .948/.950/.970 on IXI. All three policies gain above random at every fixed budget. IXI Anchor-CV oracle-normalized gain is .920/.936/.972/.973 at 10/20/30/50%; output PC is stronger at some budgets. The IXI benefit compares mean HAT/SwinIR $\times 4$ error with frozen I3Net $\times 2$ error, so it is cross-estimator and retrospective, not a same-model causal treatment effect.

Anchor-CV takes .42/3.97/3.93 ms CPU on representative fastMRI/IXI/NingXia examinations with zero neural calls. Output PC is cheap only after predictions exist. Neither Anchor-CV nor PC localizes the worst physical plane; their supported use is examination triage.

10. Material Negative and Boundary Results

1. Integer overlap is not a monotone dose; the exact analytical ladder reverses at zero-to-one sharing.
2. Fixed-count continuous influence is null/mixed on NingXia; all eight-patient intervals cross zero.
3. Local response rankings are stable to tested perturbation parameters but not invariant across physical planes.
4. The local cancellation chain is stronger than the global context-ranking chain; no causal mediation claim is supported.
5. Anchor-CV is not uniquely informative or universally superior. Adjacent differences are statistically similar and can be stronger; output PC wins several IXI comparisons.
6. Examination scores do not localize the worst plane and are not calibrated pixelwise uncertainty.
7. Densification results are retrospective, cross-estimator on IXI, and omit scanner-time, motion, and clinical utility.
8. Minimum-influence context selection transfers below random/PC on IXI and is heterogeneous on NingXia.
9. Multi-anchor jackknife IQR collapses from development to confirmation/transfer and is far below adjacent MAE.
10. Covariance residualization gives negligible IXI gain with intervals crossing zero and mixed/tied results elsewhere.

Items 8–10 test abandoned mechanism-derived methods. They are not main-paper contributions, but are retained here because they prevent the reader from inferring that the mechanism already yields an improved universal score.

11. Reproducibility and Evidence Map

The frozen sources are:

- clean intervention and stable wrong: `CVPR/CVPR_UPGRADE/outputs/clean_max_overlap*` and `stable_wrong_prevalence.json`;
- analytical ladder/influence: `compact_four_anchor_overlap_ladder*` and `shared_`

Table 5. Broad neural context grid. ρ_c uses overlap count and ρ_I neural shared influence. Advantage intervals are clustered.

Cohort	Model	ρ_c align.	ρ_I align.	Adv. [95% CI]	ρ_I shared	ρ_I observable	partial ρ_I align.
fastMRI	HAT	.799	.838	.039 [.028,.049]	.829	−.830	.359
	SwinIR	.796	.851	.055 [.048,.061]	.845	−.845	.395
IXI	HAT	.714	.797	.082 [.076,.088]	.794	−.795	.315
	SwinIR	.712	.787	.075 [.070,.080]	.786	−.787	.289
NingXia	HAT	.743	.786	.043 [.027,.057]	.792	−.792	.235
	SwinIR	.735	.772	.036 [.016,.057]	.781	−.781	.197

Table 6. Neural influence partial Spearman at exactly one shared anchor, controlling span, asymmetry, and examination. Error-alignment intervals are shown.

Cohort	Model	Error alignment	Shared fraction	Observable fraction	Disagreement
fastMRI	HAT	.169 [.053,.283]	.135	−.135	−.161
	SwinIR	.350 [.212,.464]	.321	−.321	−.286
IXI	HAT	.346 [.259,.424]	.360	−.360	−.543
	SwinIR	.258 [.181,.335]	.241	−.241	−.587
NingXia	HAT	.014 [−.232,.223]	−.006	.006	−.102
	SwinIR	−.039 [−.175,.144]	−.039	.039	−.072

`influence*`;

- neural influence and chain: `plan6_neural_shared_influence_confirm3*` and `plan6_mechanism_chain*`;
- Anchor-CV/consequence: `plan6_anchor_cv_explanation*` and `plan6_ixi_x2_consequence*`;
- Plan-7 stability and negative gates: `PLAN7_ACCEPTANCE_OPTIMIZATION/outputs/p7_cvpr_*`;
- full selection rationale: `CVPR/CVPR_RECOVERED/PAPER/EVIDENCE_SELECTION_AUDIT.md`.

Table 7. Partial Spearman ranges across HAT/SwinIR. Local arrows are supported in every stratum. "Ranking" counts architecture intervals excluding zero.

Cohort	Influence→alignment	Alignment→norm. disagreement	Influence→observable	Alignment→ranking
fastMRI	.372–.410	−.933–−.926	−.412–−.370	−.333–−.143 (0/2)
IXI	.321–.347	−.893–−.890	−.342–−.298	−.643–−.500 (2/2)
NingXia	.219–.258	−.898–−.889	−.231–−.211	−.667–−.548 (1/2)

Table 8. Examination-level $\times 4$ difficulty ranking. Lower nAURC is better.

Cohort	Method	Spearman	Kendall	nAURC
fastMRI	Anchor-CV / adjacent MAE / adjacent RMSE	.943 / .934 / .942	.812 / .790 / .804	.0231 / .0222 / .0241
IXI	Anchor-CV / adjacent MAE / adjacent RMSE	.955 / .956 / .956	.831 / .834 / .826	.0150 / .0145 / .0147
NingXia	Anchor-CV / adjacent MAE / adjacent RMSE	.958 / .973 / .959	.844 / .879 / .835	.0112 / .0072 / .0123